PalmSource™ 99

Beaming Data with the Exchange Manager

Bruce Thompson
Senior Developer Support Engineer

3Com
More connected.

Beaming Tips

- Ask before you beam
- Use the beam stroke
  - Applications—Prefs—button panel—pen
- Create a beamable business card
  - Contains only the data you want to share
  - Selected as your card for one button beaming

The Exchange Manager

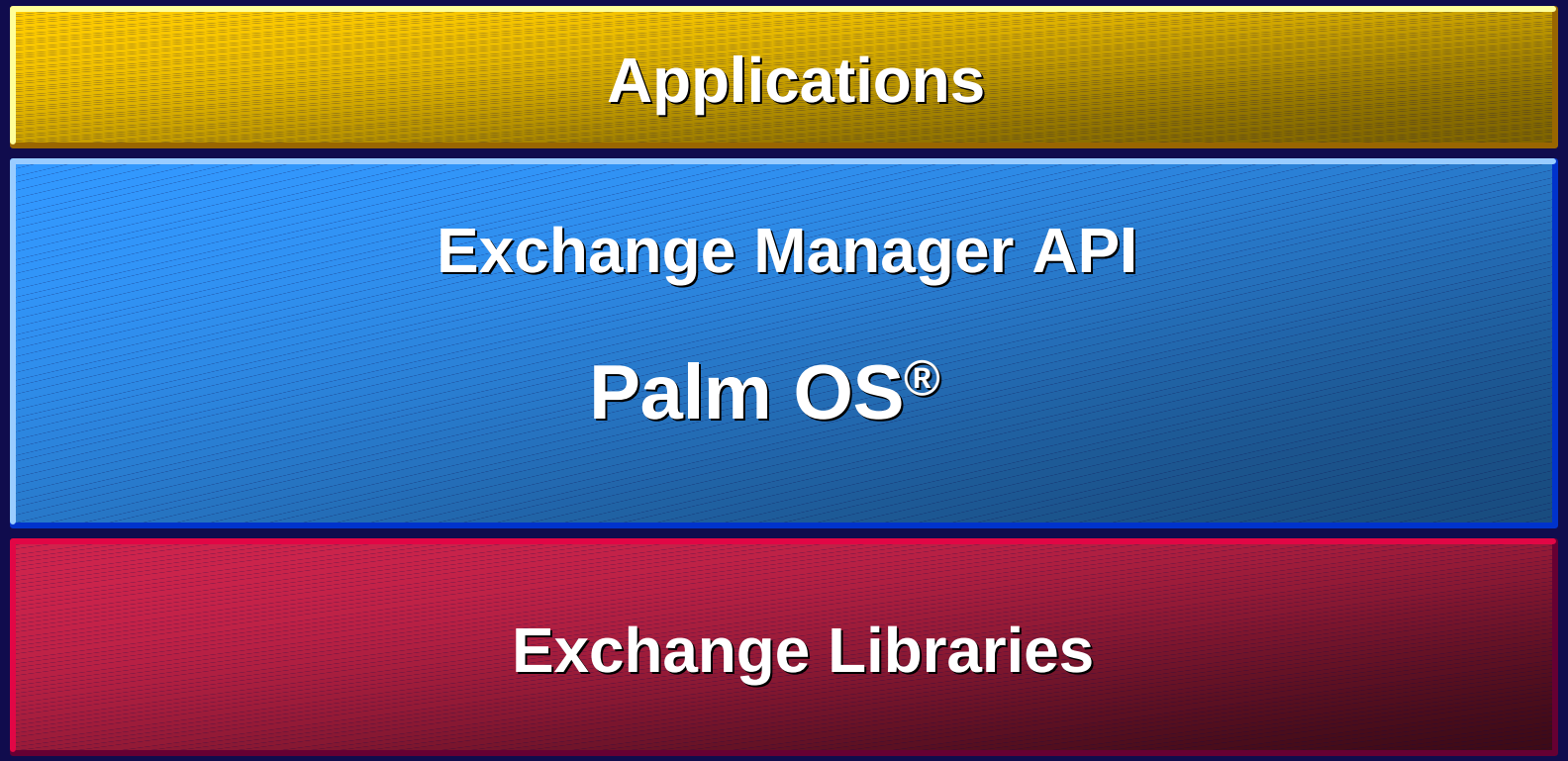
- What is the Exchange Manager?
- How should my application use it?

What Is the Exchange Manager?

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- A system that allows Palm Computing® Platform applications to exchange data directly with other devices and other applications

Exchange Manager Block Diagram



Three Goals

- Simple to use
- Standard formats
- A world of connectivity

Simple to Use

- What does it do?
- Make the import and export of data easy

The PC Way

- On Sending side
  - Export data to a file
  - Launch transfer program
  - Send the exported file

The PC Way

- On Sending side
  - Export data to a file
  - Launch transfer program
  - Send the exported file
- On Receiving side
  - Launch transfer program
  - Receive and save file
  - Open viewing app and import file

The Palm OS® Way

- On Sending side
  - Beam data

The Palm OS® Way

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- On Sending side
  - Beam data
- On Receiving side
  - Click OK

What Makes It So Simple?

- No transfer application
- No files seen by user
- No inbox
- Always ready to receive
- Opens the destination app for you

Key Point

- Each application is responsible for beaming its own data
- An application must support the Exchange Manager in order to exchange data

Standards

- Not just Palm organizer to Palm organizer, but Palm organizer to the rest of the world—directly
- Standard data formats
- Standard protocols

Standard Data Formats

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- vCard (.VCF)
- vCalendar (.VCS)
- Text (.TXT)
  - RFC 2045 quoted printable
  - RFC 2046 Multi-part MIME
- MIDI

Standard Data Formats

- Extra work now will pay off later
- Allows you to share data with other applications
- Your data can show up on non-Palm platform devices
- No Palm device is an island

Standard Protocols

- IrDA
  - Infrared Data Association
- OBEX
  - Object Exchange
- TinyTP, IrLMP, IrLAP

A World of Connectivity

- Other Palm devices
- Windows 95/98 and NT
  - QuickBeam and IrDA stack
- Cell phones, pagers and etc…
- IrMC and OBEX/vCard initiative
- Other apps on your device

More to Come…

- BlueTooth
- E-mail attachments
- Browser upload/download
- FTP and HTTP
- Pagers
- Exchange over sync

References

- Internet Mail Consortium
  - www.imc.org
- Internet Assigned Numbers Authority (MIME registry)
  - www.iana.com
- Infrared Data Association
  - www.irda.org

How Does My Application Use the Exchange Manager?

- Define an import / export format
- Add calls to send data
- Add calls to receive data

Six Functions

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- ExgRegisterData
- ExgPut
- ExgAccept
- ExgSend
- ExgReceive
- ExgDisconnect

Define an Import/Export Format

- Use existing ones when available
- Define your own
- Frees your application from version problems
- Can share data with other applications
- Define MIME type and extension

Sending Data

- Streams type interface
- Transmits single known objects
- Similar to HTTP level protocol
- Group multiple items into one object

Sending Data

- Create an ExgSocket structure
- Start send with ExgPut
- Send data with ExgSend
  - One or more times
- Disconnect with ExgDisconnect

Creating an ExgSocket

- Gives information about the data to be sent
- Allows the system to route the data to the correct application

The ExgSocketType Structure

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```
struct ExgSocketType {
    Word            libraryRef;     // Identifies the Exg library to use
    Ulong           socketRef;      // Identifies this connection
    Ulong           target;         // Creator ID of target application
    Ulong           count;          // Object count
    Ulong           length;         // Total length
    Ulong           time;           // Last modified time (optional)
    Dword           appData;        // Application specific
    Ulong           goToCreator;    // Creator ID to launch after receive
    ExgGoToType     goToParams;     // Goto find info
    Word            localMode:1;    // Exchange locally only
    Word            reserved:15;    // Set to zero always!
    CharPtr         description;    // Description of object
    CharPtr         type;           // Mime type
    CharPtr         name;           // Filename
};
```

Creating an ExgSocket

- Always zero structure first
- Filename
  - Unique filename with extension
    - For use on non-Palm platforms
    - Filename should be based on data
    - Extension should show data type
    - Example is GavinPeacock.vcf
    - Default is ExgData.exg

Creating an ExgSocket

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- Target
  - The four byte creator ID of the target Palm OS™ application
- Type
  - MIME data type (www.iana.org)
  - Example application/x-mydata
- Description
  - Shows up in “Do you want” dialog
  - Filename used is this is missing

Creating an ExgSocket

- Length
  - Zero or a hint at the size of the data
- appData
  - For your application's use
  - NOT transferred to the other machine
- Count
  - Set to 1

Creating an ExgSocket

- localMode
  - Sends to target app on the same machine

Sending Data to Another Application

- Set target creator ID
- Set local Mode flag

Sending Data

- ExgPut
- ExgSend
  - One or more times…
- ExgDisconnect

- Dialogs and errors are displayed automatically
- Check for errors on all calls

Sending Data Example


```
MemSet(&exgSocket,sizeof(exgSocket),0);
    exgSocket.description = "Beamer picture";
    exgSocket.name="Beamer.pbm";
    exgSocket.length = size;
    exgSocket.target = ‘bemr’;
    exgSocket.type = “application/x-beamer”;

    err = ExgPut(&exgSocket);   // put data to destination
    if (!err) {
          ExgSend(&exgSocket,bmpP,size,&err);
          err = ExgDisconnect(&exgSocket,err);
    };
```

Receiving Data

- Register types and extensions
- Handle Launch codes
- Accept with ExgAccept
- Receive with ExgReceive
- Disconnect with ExgDisconnect
- Go to the new data

Registering to Receive Data

- Use ExgRegisterData
- MIME type and / or filename extension
  - Separate multiples with commas
- Register on sync notification
- Automatic de-registration
  - Or register with NULL data

Handling Launch Codes

- sysAppLaunchCmdExgAskUser
  - To override ask dialog in certain cases
- sysAppLaunchCmdExgReceiveData
  - Passes ExgSocketPtr of data to receive
- sysAppLaunchCmdGoTo
  - Launched when gotoCreator is set
- sysAppLaunchCmdSyncNotify

PilotMain Example


```
Dword PilotMain(Word cmd, Ptr cmdPBP, Word launchFlags)
{
        Dword err = RomVersionCompatible(version30, launchFlags);
        if (err) return err;
    ...
        if (cmd == sysAppLaunchCmdExgReceiveData) {
                err = ReceiveData((ExgSocketPtr) cmdPBP);

        } else if  (cmd == sysAppLaunchCmdSyncNotify) {
                ExgRegisterData('bemr', exgRegExtensionID, "pbm,pb2" )
                ExgRegisterData('bemr', exgRegTypeID, "application/x-pbm");

        } else if ( cmd == sysAppLaunchCmdGoto) {
        ...
```

Accepting

- sysAppLaunchCmdExgReceiveData
  - When your application gets this launch code, call ExgAccept to begin the process of receiving data
  - Your application may not be the active program when this is called, so do not depend on global data

Receiving Data

- ExgReceiveData
  - Call one or more times to receive data
  - Returns 0 bytes when last of the data is read
  - Launched when gotoCreator is set
- ExgDisconnect
  - Pass any error your app had saving the data

Displaying Data

- gotoCreator
  - Set to creator ID of app to launch with goto launchcode
  - Set to zero to not goto
- gotoParams
  - Fields are copied into a standard goto structure
  - Matchcustom is for your apps own use
    - pass uniqueIDs in matchcustom

The ExgGotoType Structure

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```
typedef struct {
    Word        dbCardNo;       // card number of the database
    LocalID     dbID;           // Local ID of the database
    Word        recordNum;      // index of record that contain a match
    Dword       uniqueID;       // position in record of the match.
    Dword       matchCustom;    // application specific information
} ExgGoToType;
```

Receiving Data Example


```
err = ExgAccept(&exgSocket);   // put data to destination
if (!err) {
        err = ExgReceive(&exgSocket,bmpP,size,&err);
        err = ExgDisconnect(&exgSocket,err);
}
if (!err) {
    exgSocketP->gotoCreator = 'bemr';
    exgSocketP->gotoParams.matchCustom = (Dword)          dataH;
};
```

Sending Database Files

- Converts between .prc/.pdb file formats and internal formats
  - ExgSendDatabase
    - Use to send whole databases
    - Will overwrite receivers copy
  - ExgReceiveDatabase
    - Only used by launcher replacements

Debugging Techniques

- Use Test (loop back) mode on device while debugging
- Use Gremlins in Mac simulator
- Not working in Emulator yet
- Beam files to / from desktop
  - Use Quickbeam application
  - Direct Serial connection is possible

Tools

- Quickbeam application
  - www.countersys.com
- Ir port for desktops
  - www.actisys.com and others

Magic Keystrokes

- Shortcut .t (Test Mode)
  - Loop back to your own app
- Shortcut .s (Serial Mode)
  - User serial cable instead of Ir
- Shortcut .i (connect Now)
  - To receive when beam receive is off

Summary

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- The Exchange Manager
- A new world of communications
- Make your application a part of it

Q&A


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